

Ahmed Loughzali

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 [LinkedIn](#)

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Education

Fall 2026 **Incoming Exchange Student.**

Gwangju Institute of Science and Technology (GIST), South Korea

Focusing on Robotics and Physical AI. Selected Coursework: Multiple View Geometry for Spatial AI, Robotics, Haptics, AI for Healthcare, Biomechanics.

2022 – 2027 **MSc in Computer Science.**

IMT Nord Europe, France

Majoring in AI & Data Science.

Experience

May 2026 – Aug. **AI Researcher.**

2026 *Algoverse, Remote*

Selected for a competitive research program (scholarship + compute grant). Researching the adversarial robustness of discrete image tokenizers in medical imaging pipelines

Apr. 2026 – Sept. **AI & Robotics Research Intern.**

2026 *Enchanted Tools, Paris, France*

- Working on the brain of Mirokaï, Europe's most advanced social humanoid robot: on the conversational audio stack (turn-taking, user barge-in, semantic VAD).
- Implementing and benchmarking state-of-the-art turn-taking models (VAP, TurnGPT, SmartTurn, Semantic-VAD via Silero & custom Whisper); fine-tuning Qwen-based models for multilingual end-of-turn detection.
- Built an end-to-end pipeline for barge-in detection: synthetic data generation on NVIDIA Thor, model training, and real-time deployment on Jetson Orin NX.
- Optimizing models for real-time human-robot interaction under strict latency constraints (quantization), and streamlining the edge deployment lifecycle: cross-compilation, gRPC service integration, and CI/CD automation via GitLab pipelines.
- Selected to represent the company and pitch the robot's technical capabilities to enterprise customers at major industry events (Microsoft stand at VivaTech and Santexpo).

May 2025 – Aug. **AI Engineer Intern.**

2025 *Ferromobile, Paris, France*

- Built and optimized a real-time Text-to-Speech service (FastAPI, Piper TTS) on NVIDIA Jetson Xavier, achieving ~0.6s latency with caching, integrated via TCP/HTTP to deliver voice guidance to passengers of the autonomous vehicle.
- Designed and deployed a local RAG pipeline (Docling, LangChain, ChromaDB, Mistral 7B via Ollama) with a Streamlit web interface for intelligent querying of technical documents.
- Containerized the full architecture using Docker Compose and Dev Containers to ensure reproducible and reliable deployment on embedded industrial hardware.

Projects & Research

- June 2026 **Selected (Top 15%, 100/650), Hack the World(s).**
World Models Hackathon, endorsed by Yann LeCun
 Built a surgical world model: an action-conditioned JEPA (EB-JEPA) predicting future latent states from wrist-camera video and proprioception.
- Feb. 2026 **1st Place Winner, Mistral AI x GE Healthcare Hackathon.**
OncovIA
 Developed a multimodal pipeline for oncology combining Computer Vision and Agentic RAG to help radiologists generate accurate, patient-specific follow-up reports.
- Oct. 2025 – Feb. 2026 **Drone Detection using Spiking Neural Networks.**
Research Project
 Authored a review paper on drone detection focusing on neuromorphic computing and energy-efficient SNNs. Fine-tuned spiking neural network architectures (Spiking YOLOv8, Spiking SSD) on the FRED dataset. Achieved 92% accuracy with a 69M parameter model optimized for low-power edge inference.
- Sep. 2024 – Apr. 2025 **Ozone Measurement System on Drone.**
Embedded Hardware & IoT Project
 Designed and built a low-cost (< 150€) autonomous embedded system mounted on a drone for real-time ozone measurement (ESP8266, SEN0321, GPS). Handled full hardware design, wiring, C++ embedded software, and remote server networking (Flask + InfluxDB + Grafana), achieving an 80% correlation with a 3000€ professional reference sensor.

Technical Strengths

Languages	Python, C++, SQL, Bash, LaTeX
Frameworks	PyTorch, Hugging Face, LangChain, FastAPI, gRPC, OpenCV, NumPy, Pandas, Scikit-learn
Tools & Infra	Linux, Git, Docker, GitLab CI/CD
Hardware & Edge	NVIDIA Jetson, ESP, Cross-Compilation, Low-Power Inference

Languages

English	Full Professional Proficiency	TOEIC: 980/990
French	Mother tongue	
Arabic	Mother tongue	
Spanish	Intermediate	
Russian	Beginner	

Interests

Sports	Boxing, Wrestling, Hiking.
Other	Reading.